No new Palpatine
Star Wars actor Ian McDiarmid who played Emperor Palpatine, doesn't want anyone else to play the character. **http://dgit.in/EmprrPlptn**

More rumours about Gambit
Stuck in development for a really long time, new rumours and plot details are making their way. **http://dgit.in/GambitMv**

THE DIGIT COMMUNITY IS FILLED WITH BRIGHT IDEAS and we find the best among the best every month to bring them to you in this section.

Cold Fusion: could it be the answer to our energy woes?

Being a long standing joke, cold fusion is finally being taken seriously. We explore why.

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A member in the list of alternative energy source since the 90's, cold fusion promises to be a clean, cheap and possibly limitless source of energy. Once considered to be a hoax (pathological science), it has gained a lot of traction in recent years with even commercial products surfacing.

WHAT IS COLD FUSION?

A nuclear reaction is any reaction happening at the atomic level. There are two types of nuclear reactions, fission and fusion. Fission is where a heavier atom splits into lighter atoms, releasing energy. For example, in nuclear power plants, Uranium is used as fuel which splits into Barium and Krypton. On the other hand, fusion is where two lighter atoms combine to form a heavier atom with extra mass becoming energy. Our sun is powered by fusion reaction where two or more Hydrogen atoms combine to form Helium.

Fusion is better than fission for a number of reasons. The major one is the waste products released. In a fission reaction, the waste products are highly radioactive and difficult to dispose off safely, whereas in fusion, the waste product is just plain old helium. With all of these benefits going for fusion, why hasn't it become mainstream? A lot of energy is required to fuse atoms. Uranium is naturally unstable and is begging to be split. Hydrogen on the other hand, is very stable and needs some coaxing to fuse.

Enter cold fusion, a hypothetical nuclear fusion reaction that occurs at room temperature. The setup for these reactions are also far more simpler than the complex tokamaks required to run traditional fusion reactions. A simplified cold fusion setup can be abstracted to dipping two electrodes into a solution of heavy water and Palladium, similar to an electrolysis setup. Researchers claim to have observed hydrogen fusing at room temperature and releasing heat that cannot be explained by the current level of electrochemistry. If these claims prove true, unlimited energy may not be restricted only to the realm of science fiction.

Martin Fleischmann and Stanley Pons with their electrolysis tube.

A BRIEF HISTORY OF COLD FUSION

The story of cold fusion begins in 1989, when two scientists from Utah, Martin Fleischmann and Stanley Pons claimed to have achieved nuclear fusion in an electrolysis experiment. The published results caused an uproar at that time. Over the next few weeks, countless researchers tried to replicate the results, but to no avail. The inability to replicate, led to the mainstream scientific community to dismiss the experiment as a hoax and brand cold fusion research as junk. Scientists who pursued cold fusion research were ridiculed, mainly by the remainder of the community. As a result, cold fusion died a cold death. However, despite the mainstream view, a small percentage of determined researchers continued to experiment on cold fusion.

THE WORLD FINALLY WARMS UP TO COLD FUSION

It's called LENR (Low-energy nuclear reaction) now to remove the toxicity attached to the former name. The name change also suggests the broader possibility that what's happening under the hood is not fusion but some other unidentified nuclear reaction. The modern world's shift in perception towards LENR can be marked by the interest